

Multiple Choice

1. **Answer:** A

Options: A) 5.4 moles per mile; B) 9.0 moles per mile; C) 2.5 moles per mile; D) 7.3 moles per mile; E) 3.8 moles per mile

Note: Correct option: A - 5.4 moles per mile

2. **Answer:** A

Options: A) 350 lbs. sulfur; B) 320 lbs. sulfur; C) 425 lbs. sulfur; D) 400 lbs. sulfur; E) 300 lbs. sulfur

Note: Correct option: A - 350 lbs. sulfur

3. **Answer:** D

Options: A) 8500 K; B) 15000 K; C) 9500 K; D) 11000 K

Note: Correct option: D - 11000 K

4. **Answer:** A

Options: A) 43 minutes; B) 103 minutes; C) 93 minutes; D) 113 minutes; E) 23 minutes

Note: Correct option: A - 43 minutes

5. **Answer:** A

Options: A) C₃ H₆; B) C₅ H₁₀; C) C₄ H₈; D) C₄ H₉; E) C₃ H₈

Note: Correct option: A - C₃ H₆

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '35 M H₂ SO₄'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1.8610²⁸'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '60 g'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '4.2 10⁻²³ kg · m · s⁻¹'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Photon wavelength = 1300 nm, Electron wavelength = 1.300 nm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1.10×10^3 oz. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key